

解樂端 112700026

18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$$H_0: p_1 = p_2 = p_3 = \frac{1}{3}$$

H_a : not all proportions are equal

Entrance	1	2	3	Total
Observed (O_i)	83	61	56	200
Expected (E_i)	$np_i = 66.67$	$np_i = 66.67$	$np_i = 66.67$	200
$\frac{(O_i - E_i)^2}{E_i}$	4.00	0.48	1.71	6.19

$$df = k - 1 = 2$$

$$\alpha = 0.05$$

$$\chi^2_{2, 0.05} = 5.991$$

$$\therefore 6.19 > 5.991$$

$\therefore$ Reject H_0

$$\begin{aligned}
 95\% \text{ C.I. for Entrance 1} &= \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \\
 &= 0.415 \pm z_{0.025} \sqrt{\frac{0.415(0.585)}{200}} \\
 &= 0.415 \pm 1.96(0.0348) \\
 &= 0.415 \pm 0.068 \\
 &= (0.347, 0.483)
 \end{aligned}$$

$\therefore$ 95% C.I. for Entrance 1 is (0.347, 0.483)

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.¹ In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$$H_0: p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = p_7$$

$$H_a: \text{not all proportions are equal}$$

Day	Sun	Mon	Tue	Wed	Thur	Fri	Sat	Total
O_i	24	36	27	26	32	26	29	200
E_i	$\frac{200}{7} \approx 28.57$	28.57	28.57	28.57	28.57	28.57	28.57	200
$\frac{O_i - E_i}{E_i}$	0.73	1.93	0.09	0.23	0.41	0.23	0.01	3.63

$$df = k - 1 = 6$$

$$\alpha = 0.05$$

$$\chi^2_{6,0.05} = 12.592$$

$$\therefore 3.63 < 12.592$$

$\therefore$ Non-reject H_0 , there is not sufficient evidence to indicate that we think the incidence of heart attacks differs depending on the day of the week.

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large
Fatal	67	26	16
Not Fatal	128	63	46

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

H_0 : Fatal accident frequency is independent of automobile size
 H_a : Fatal accident frequency is dependent on automobile size

	Small	Medium	Large	Row Total
Fatal	67	26	16	109
Not Fatal	128	63	46	237
Column Total	195	89	62	346

Fatal & Small

$$E_{ij} = \frac{109(195)}{346} = 61.43$$

Not Fatal & Small

$$E = \frac{237(195)}{346} = 133.57$$

Fatal & Medium

$$E_{ij} = \frac{109(89)}{346} = 28.04$$

Not Fatal & Medium

$$E = \frac{237(89)}{346} = 60.96$$

Fatal & Large

$$E_{ij} = \frac{109(62)}{346} = 19.53$$

Not Fatal & Large

$$E = \frac{237(62)}{346} = 42.47$$

	Small	Medium	Large	Total
Fatal Contribution	$\frac{(67-61.43)^2}{61.43} = 0.505$	$\frac{(26-28.04)^2}{28.04} = 0.148$	$\frac{(16-19.53)^2}{19.53} = 0.638$	1.291
Not Fatal Contribution	$\frac{(128-133.57)^2}{133.57} = 0.232$	$\frac{(63-60.96)^2}{60.96} = 0.068$	$\frac{(46-42.47)^2}{42.47} = 0.293$	0.593
				1.884

$$df = (2-1)(3-1) = 2, \chi^2_{2,0.05} = 5.991$$

$$\therefore 1.884 < 5.991$$

$\therefore$ Non-reject H_0

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

Salary	Workdays at Home per Week		
	Less Than One	At Least One, but Not All	All at Home
Under \$50,000	38	16	14
\$50,000–\$74,999	54	26	12
\$75,000–\$99,999	35	22	9
Above \$100,000	33	29	12

- Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.
- Use Table 5 in Appendix I to approximate the p -value for this test of hypothesis. Does the p -value confirm your conclusions from part a?

H_0 : Salary and number of workdays at home are independent
 H_a : Salary and number of workdays at home are dependent

Salary	Workdays at Home per Week			
	Less Than One	At Least One, but Not All	All At Home	Row Total
Under \$50,000	38	16	14	68
\$50,000 – \$74,999	54	26	12	92
\$75,000 – \$99,999	35	22	9	66
Above \$100,000	33	29	12	74
Column Total	160	93	47	300

Expected Table

Salary	Less Than One	At Least One, but Not All	All At Home
Under \$50,000	$\frac{68(160)}{300} = 36.27$	$\frac{68(93)}{300} = 21.08$	$\frac{68(47)}{300} = 10.65$
\$50,000 – \$74,999	$\frac{92(160)}{300} = 49.07$	$\frac{92(93)}{300} = 28.52$	$\frac{92(47)}{300} = 14.41$
\$75,000 – \$99,999	$\frac{66(160)}{300} = 35.2$	$\frac{66(93)}{300} = 20.46$	$\frac{66(47)}{300} = 10.34$
Above \$100,000	$\frac{74(160)}{300} = 39.47$	$\frac{74(93)}{300} = 22.94$	$\frac{74(47)}{300} = 11.59$

Salary	Less than One	At least One, but Not All	All At Home	Total
Under \$50,000	$\frac{(38-36.27)^2}{36.27} = 0.083$	$\frac{(16-21.08)^2}{21.08} = 1.226$	$\frac{(14-10.65)^2}{10.65} = 1.055$	2.364
\$50,000 - \$74,999	$\frac{(54-49.07)^2}{49.07} = 0.495$	$\frac{(26-28.52)^2}{28.52} = 0.223$	$\frac{(12-14.41)^2}{14.41} = 0.403$	1.121
\$75,000 - \$99,999	$\frac{(35-35.2)^2}{35.2} = 0.001$	$\frac{(22-20.46)^2}{20.46} = 0.116$	$\frac{(9-10.34)^2}{10.34} = 0.174$	0.291
Above \$100,000	$\frac{(33-39.47)^2}{39.47} = 1.061$	$\frac{(29-22.94)^2}{22.94} = 1.601$	$\frac{(12-11.89)^2}{11.89} = 0.015$	2.677
				6.453

$$df = (4-1)(3-1) = 6$$

$$\chi^2_{6,0.05} = 12.592$$

$$\therefore 6.453 < 12.592$$

$\therefore$ Non-reject H_0 , there is not sufficient evidence to indicate that salary is dependent on the number of workdays spent at home.

$$(b) \chi^2 = 6.453$$

$$df = 6$$

$$\chi^2_{6,0.50} = 5.348$$

$$\chi^2_{6,0.25} = 7.841$$

$$\therefore 5.348 < 6.453 < 7.841$$

$$\therefore 0.25 < p\text{-value} < 0.50$$

$$\therefore p\text{-value} > 0.05$$

$\therefore$ Non-reject H_0